

EE2213 Introduction to AI

Lecture 7: Introduction to Machine Learning

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OVERVIEW OF COURSE CONTENTS

- **Introduction (Shaojing)**

- What is AI
- Applications of AI
- AI agent

- **Search (Shaojing)**

- Uninformed search algorithms: breadth-first, depth-first, uniform-cost(Dijkstra's algorithm)
- Informed search algorithms: greedy best-first, A*
- Applications

- **Optimisation (Shaojing)**

- Linear programming
- Convex problems
- Applications

- **Machine learning (Wang Si) (Weeks 6-9)**

- Supervised and unsupervised learning: regression, classification, clustering
- Neural networks and deep learning
- Applications

- **Knowledge representation (Wang Si) (Weeks 10-11)**

- Knowledge Representation and Reasoning
- Propositional Logic
- Applications

- **Ethical considerations (Shaojing)**

- Bias in AI
- Privacy concerns
- Societal impact

(Week 10, Monday)

**No Class on 20 Oct.! A recording
will be uploaded on Canvas.**

Reference Books for Si's Part

Main:

- Stuart J. Russell and Peter Norvig, “*Artificial Intelligence: A Modern Approach-4th Edition*”

Supplementary:

- James, Gareth, et al, “*An Introduction to Statistical Learning*”. Vol. 112. New York: springer, 2013. <http://www-bcf.usc.edu/~gareth/ISL/>
- Andriy Burkov, “*The Hundred-Page Machine Learning Book*”, 2019

Continuous Assessments for Si's Part

✓ 2 In-classe Quizzes

Week	Date	CAs	%	Coverage
9	Oct. 17	Quiz 3 (1pm-1:30pm)	5%	Weeks 6-9 (Lectures & Tutorials)
11	Oct. 31	Quiz 4 (1pm-1:30pm)	5%	Weeks 10-11 (Lectures & Tutorials)

- Mode: Digital exam conducted via Exemplify. **Please bring your own computer for the quiz as you may need code to get the answers.**
- Open-book exam: You may bring any printed or digital materials, but internet access is not allowed.

✓ 1 Individual Project (Machine Learning, 20%)

Release Date: **Sept. 29** (Week 7, Monday)

Due Date: **Oct. 27** (Week 11, Monday)

✓ 1 Assignment (Propositional Logic, 5%)

Release Date: **Oct. 24** (Week 10, Friday)

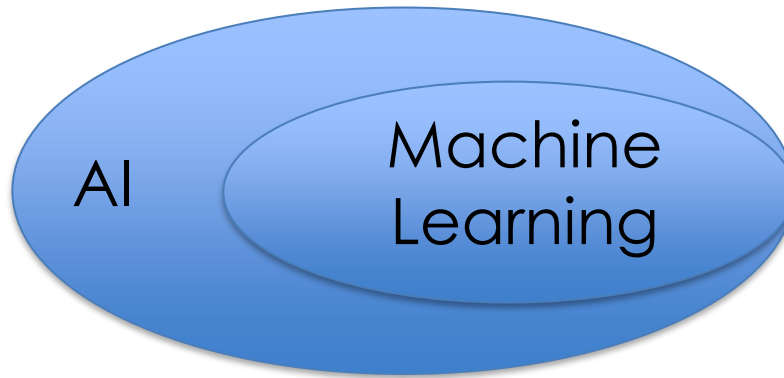
Due Date: **Nov. 7** (Week 12, Friday)

AGENDA

- We will discuss:
 - What is machine learning?
 - Types of machine learning
 - General workflow of machine learning

At the end of this lecture, you should be able to:

- ✓ Describe the 3 key components of machine learning
- ✓ Recognize different types of machine learning
- ✓ Know the general pipeline of applying machine learning



What is Machine Learning?

AI Techniques that give computers **the ability to learn** without being explicitly programmed to do so

- Arthur Samuel



What is Machine Learning?

A computer program is said to learn if its performance at task **T**, as measured by **P**, **improves** with **experience** **E**.

- Tom Mitchell

e.g., Google Maps finding the fastest route based on current traffic?

Google Maps predicting traffic jams based on historical traffic patterns?

millions of past trips from
other users

Experience
(**E**)

traffic jam
prediction

Task
(**T**)

Prediction accuracy

Performance
Measure
(**P**)

What is Machine Learning?

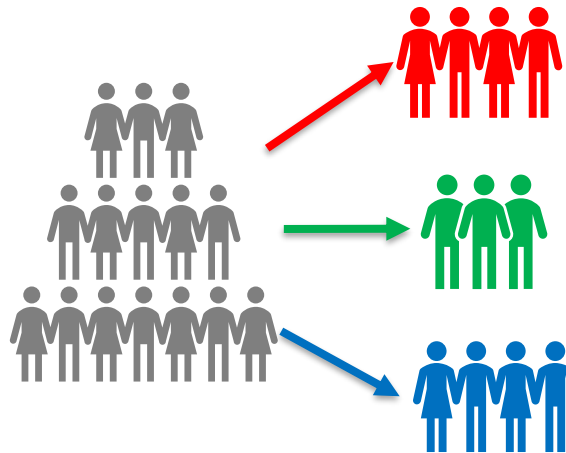
Examples: Task (**T**) Experience (**E**) Performance (**P**)



T: Spam Email Detection

E: Email Data

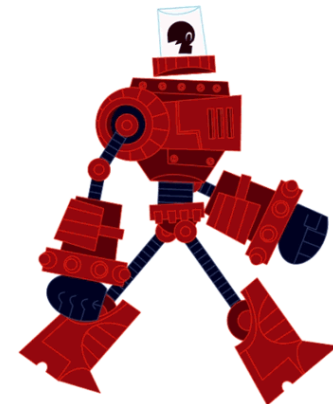
P: Detection Accuracy



T: Customer Segmentation

E: Purchase data

P: Silhouette Scores



T: Robot Walking

E: Past records of Walking

P: Walking time

Types of Machine Learning

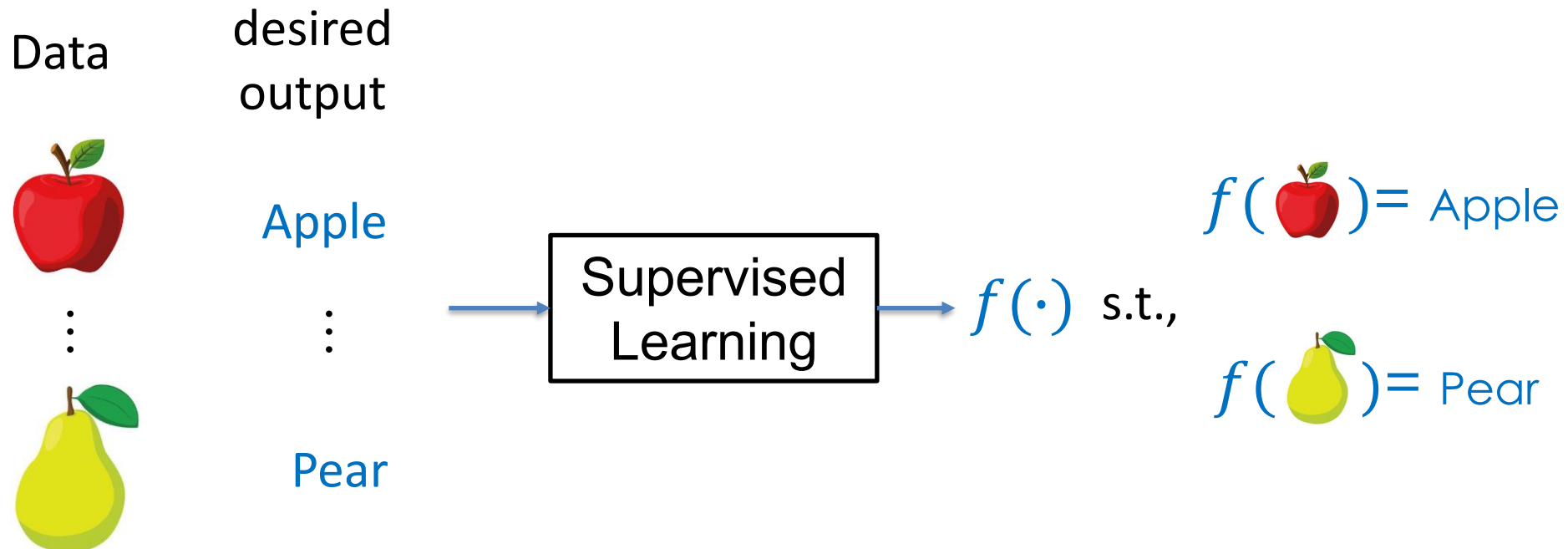
- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Types of Machine Learning

➤ Supervised Learning

Input: data with desired output

Output: A function(i.e., model) that maps data to the desired output



Types of Machine Learning

➤ Supervised Learning

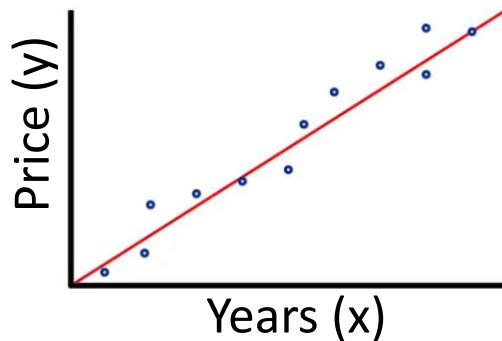
Input: data with desired output

Output: A function (i.e., model) that maps data to the desired output



Regression

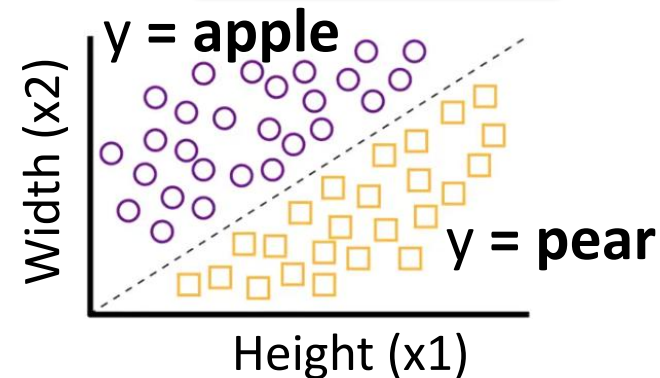
Lecture 8



*find the line that **best aligns** with samples*

Lecture 9

Classification



*find the line that **separates** two classes*

Types of Machine Learning

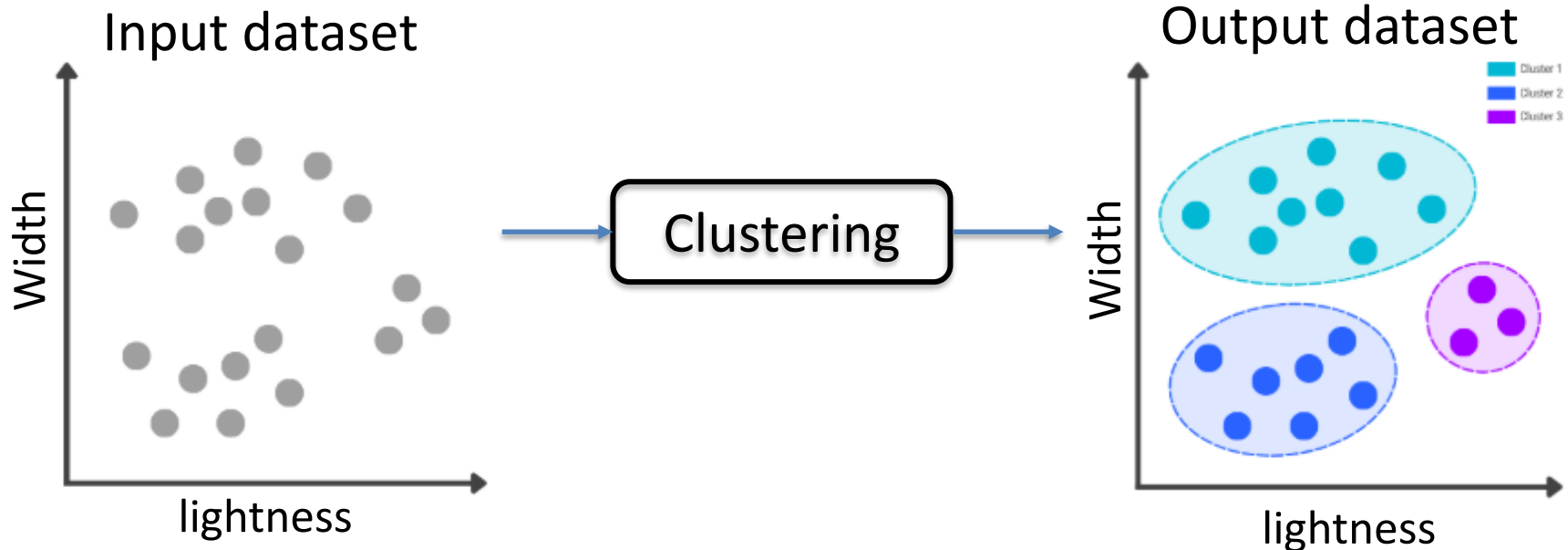
➤ Unsupervised Learning

Input: data (No desired output)

Output: Underlying patterns in data

Clustering

Lecture 10

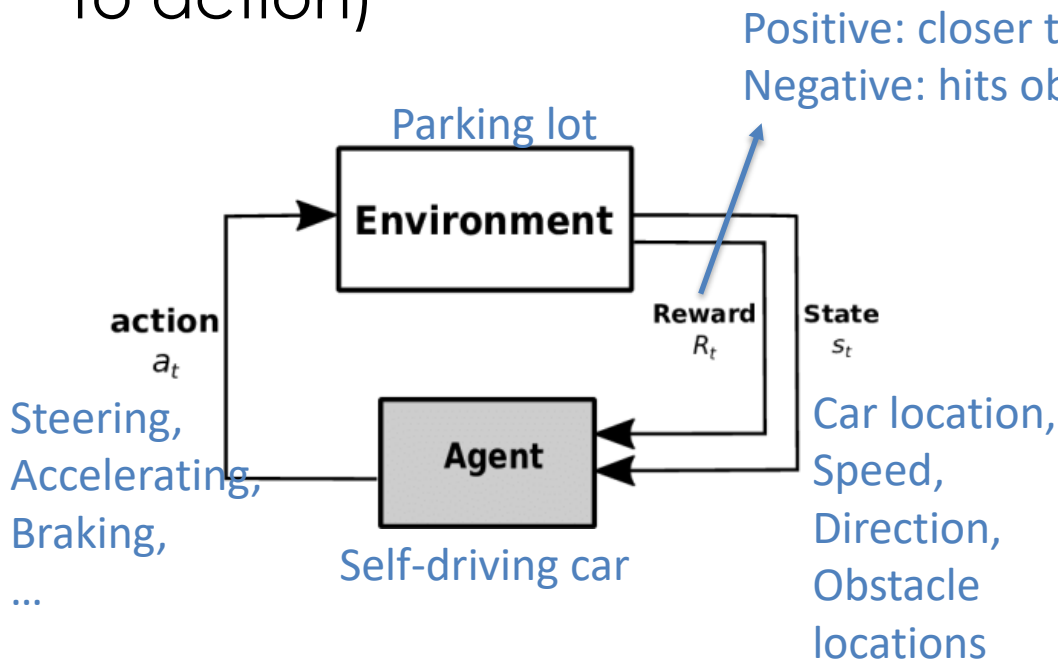


Types of Machine Learning

➤ Reinforcement Learning

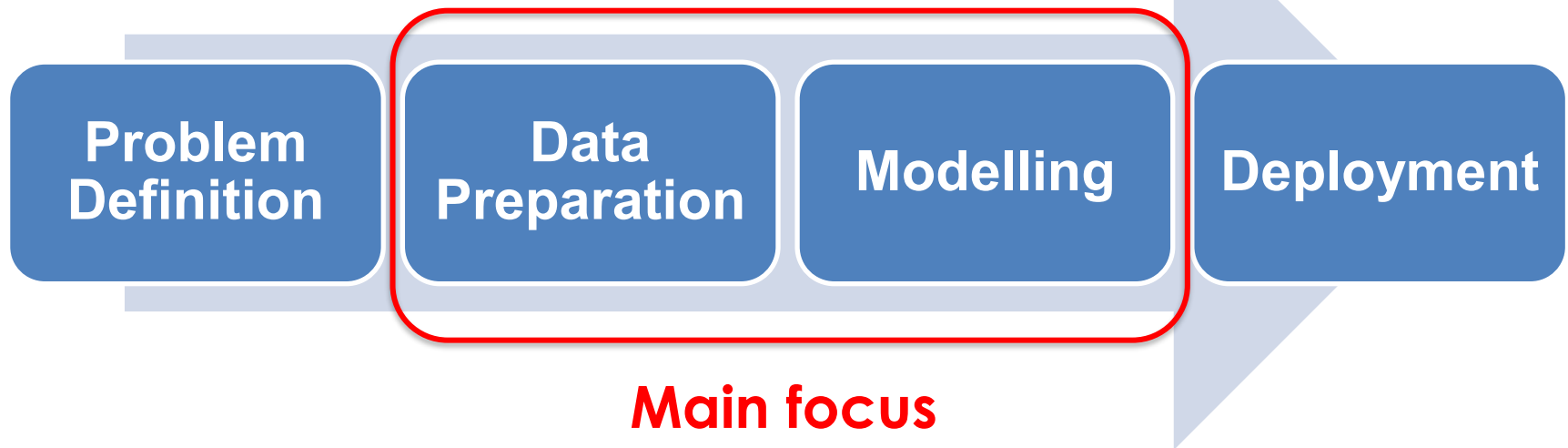
Input: Sequence of States, Actions, and Delayed Rewards

Output: Action Strategy (a rule that maps the environment to action)



Self-driving car learn to park

Workflow of Machine Learning



Workflow of Machine Learning

Problem Definition

Data Preparation

Modelling

Deployment

Example

1. Clarify the objective

What problem are you trying to solve?

Predict House Prices

2. Determine if ML is appropriate

Can this problem be learned from data?

Yes

3. Specify inputs and outputs

What input features will the model use?

Size, location, no. rooms, etc.

What is the model trying to predict?

Sale price in SGD

4. Choose the task type

Classification? Regression? Clustering?...

Regression

5. Decide on performance measure

Accuracy? Mean Squared Error?...

Mean Squared Error

Workflow of Machine Learning

Problem Definition

Data Preparation

Modelling

Deployment

1. Data Collection

Gather raw data from sensors, databases, files, etc.

2. Exploratory Data Analysis

Flag any anomalies and inconsistencies.

e.g., Predict House Prices

No.	Size	No. rooms	Location	Sold Price
1	Features			Desired output
...				
100				

Missing value

No.	Size	No. rooms	Location	Sold Price
1	112	2	"Pioneer"	600000
2		4	"Clementi"	801256
3	100	3	4	"650K"
4	116	3	"City Hall"	2300000
...				
100	112	2	"Pioneer"	600000

Inconsistent data values

Outlier

Duplicate data

Workflow of Machine Learning

Problem Definition

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Deployment

3. Data Preprocessing

- Address the problems flagged in the previous step.

e.g., handling missing value through **dropping** or **imputation**

If the number of samples is large.

Option 1: take the average value of the corresponding feature.

Option 2: take a value outside the normal range.

Missing
value

No.	Size	No. rooms	Location	Sold Price
1	112	2	"Pioneer"	600000
2		4	"Clementi"	801256
3	100	3	4	"650K"
4	116	3	"City Hall"	1300000
...				
100	112	2	"Pioneer"	600000

Workflow of Machine Learning

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3. Data Preprocessing

- Format the data

- ❖ Convert non-numerical data into numerical form.

e.g., Use postal code for location

Convert categories into a binary vector through **one-hot encoding**.

Categories: ['Cat', 'Dog', 'Bird'] $\xrightarrow{\text{one-hot encoding}}$ Cat: [1,0,0]
Dog: [0,1,0]
Bird: [0,0,1]

Handwritten: 0 1 2

- ❖ **Normalization (Ensure Features Contribute Fairly)**

Across Samples for Each Feature

Use only
training
data
Statistics!

Min-max normalization: $x_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$

Z-score standardization: $x_{scaled} = \frac{x - \mu}{\sigma}$ $\rightarrow$ Mean
 $\rightarrow$ Standard deviation

No.	Feature 1	Feature 2
1	3 <i>2</i>	-1 <i>1-1</i>
2	2 <i>min 0</i>	0 <i>2-0</i>
3	4 <i>max 4</i>	1 <i>1</i>

Handwritten calculations:
For Feature 1: $\frac{3-2}{4-2} = \frac{1}{2}$
For Feature 2: $\frac{1-(-1)}{1-(-1)} = \frac{2}{2} = 1$

Workflow of Machine Learning

Problem Definition

Data Preparation

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Deployment

4. Feature Selection or Feature Extraction

Manipulating existing features to obtain **more relevant** features that improves performance

Feature Selection	Feature Extraction
Chooses from existing features	Creates new features from original ones
e.g., drop “Favorite color”	e.g., combine “height” and “weight” into “BMI”

Example task: Predict health risk

	Height (m)	Weight (kg)	Age	Gender	Favorite color	Risk
1	1.75	59	33	Male	Red	High
...	...	...	...	...	...	...
100	1.65	50	31	Female	Blue	Medium

Workflow of Machine Learning

Problem Definition

Data Preparation

Modelling

Deployment

4. Feature Selection and Feature Extraction

Manipulating existing features to obtain **more relevant** features that improves performance

-Feature Selection with **Pearson's correlation coefficient (r)**

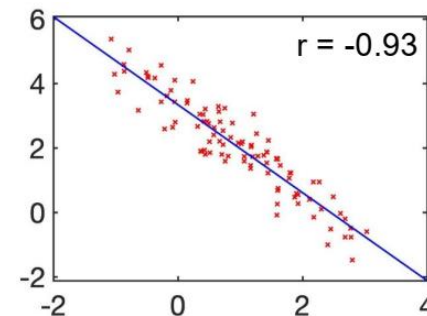
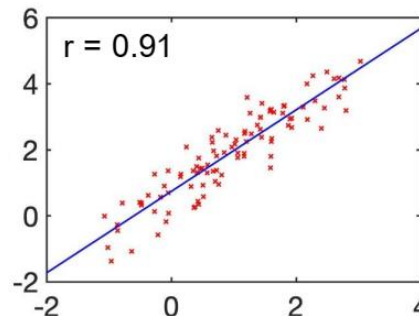
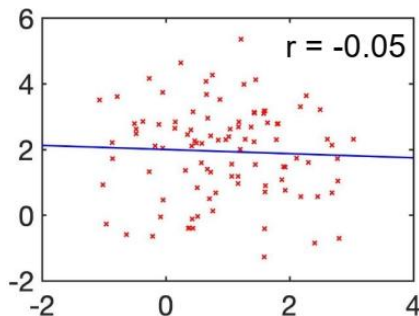
(Measures linear relationship between two numerical variables)

Given N pairs of data points $\{(a_1, b_1), \dots, (a_N, b_N)\}$

Use only training data!

$$r = \frac{\frac{1}{N} \sum_{i=1}^N (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\frac{1}{N} \sum_{i=1}^N (a_i - \bar{a})^2} \sqrt{\frac{1}{N} \sum_{i=1}^N (b_i - \bar{b})^2}}$$

where $\bar{a} = \frac{1}{N} \sum_{i=1}^N a_i$, $\bar{b} = \frac{1}{N} \sum_{i=1}^N b_i$



Workflow of Machine Learning

Problem Definition

Data Preparation

Modelling

Deployment

4. Feature Selection and Feature Extraction

Manipulating existing features to obtain **more relevant** features that improves performance

- Feature Selection with **Pearson's correlation coefficient (r)**
(Measures linear relationship between two numerical variables)

Way 1:

- Compute r between each input feature and the target output in the **training set**.
- Select features with high absolute r .

Way 2:

- Compute r between every pair of input features in the **training set**
- Remove features with high absolute r .

Workflow of Machine Learning

Problem Definition

Data Preparation

Modelling

Deployment

5. Split data into training, validation and test sets

Training

For training the ML models
(learn internal parameters)

Validation

For choosing the
hyperparameter
of the model

Test

For testing the
“real”
performance
and
generalization

Seen/visible data

Unseen/hidden
data

No.	Size	No. rooms	Location	Sold Price
1				
...		Training Set		
80				

No.	Size	No. rooms	Location	Sold Price
81				
...		Validation Set		
90				

No.	Size	No. rooms	Location	Sold Price
91				
...		Test Set		
100				

Workflow of Machine Learning

Problem Definition

Data Preparation

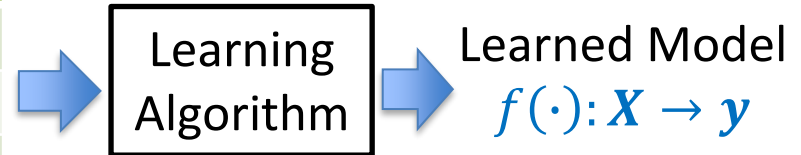
Modelling

Deployment

1. Train model on the training set

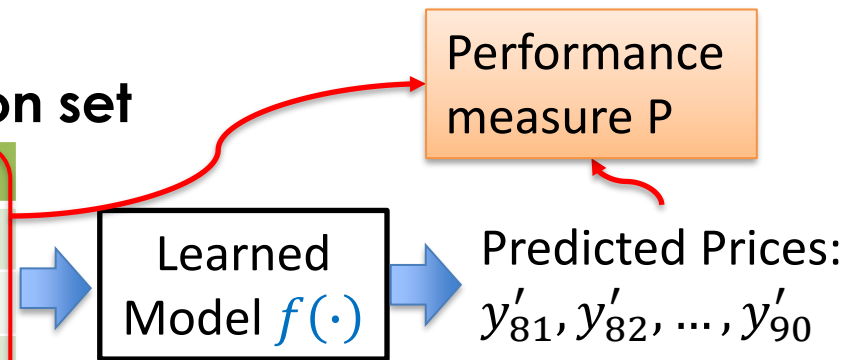
	No.	Size	No. rooms	Location	Sold Price
x_1	1	$x_{1,1}$	$x_{1,2}$	$x_{1,3}$	y_1
	...				
x_{80}	80	$x_{80,1}$	$x_{80,2}$	$x_{80,3}$	y_{80}

$\{(x_1, y_1), (x_2, y_2), \dots, (x_{80}, y_{80})\}$



2. Evaluate the model on the validation set

	No.	Size	No. rooms	Location	Sold Price
x_{81}	81	$x_{81,1}$	$x_{81,2}$	$x_{81,3}$	y_{81}
	...				
x_{90}	90	$x_{90,1}$	$x_{90,2}$	$x_{90,3}$	y_{90}



Workflow of Machine Learning

Problem Definition

Data Preparation

Modelling

Deployment

3. Hyperparameter Tuning

Change the hyperparameters and repeat steps 1&2 to find the best hyperparameters that have the **best validation performance**.

	Hyperparameters	Parameters
Definition	External settings chosen before training	Internal variables learned from the data
Learned during the training?	No	Yes
Set by user?	Yes	No
Examples	-Learning rate -Training steps	-Regression Coefficients in Linear Regression

4. Test model on the test set

Unbiased evaluation of the final model

Workflow of Machine Learning

Problem Definition

Data Preparation

Modelling

Deployment

- **Model Deployment (real-world environment)**

Web API : Model exposed as an endpoint that APPs can call

Cloud: AWS, GCP, or Azure provide managed ML serving platforms.

Edge: Model runs locally on devices like smartphones, IoT devices, etc.

Summary by Quick Quiz

- ✓ Three Components in ML Definition
- ✓ Three Types of ML
- ✓ Three ways of Handling Missing Value
- ✓ Data Formatting
 - Categorical data? Different Scales for Features?
- ✓ One Criterion for Feature Selection
- ✓ Dataset Partition

Review of Fundamentals

Review of Fundamentals

❖ Scalars, Vectors, Matrices

- A **scalar** is a simple numerical value (e.g., 10, -4.2,).
Denoted by an *italic* letter (e.g., x , y)

- A **vector** is an ordered list of scalar values.
Denoted by a **bold** letter (e.g., $\mathbf{x}$, $\mathbf{y}$)

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- A **matrix** is a rectangular array of numbers arranged in rows and columns

Denoted by a **bold** CAPITAL letter (e.g., $\mathbf{X}$, $\mathbf{Y}$)

$$\mathbf{X} = \begin{bmatrix} x_{1,1} & x_{1,2} & x_{1,3} \\ x_{2,1} & x_{2,2} & x_{2,3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1.5 \\ 2 & 9 & -3.2 \end{bmatrix}$$

Review of Fundamentals

❖ Operations on Vectors and Matrices

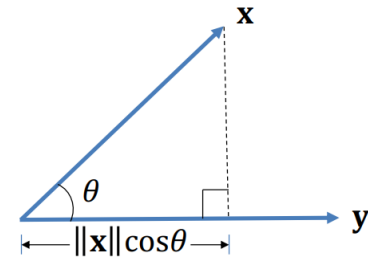
$$\mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{x}^T = [x_1 \quad x_2]$$

$$\mathbf{x} - \mathbf{y} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 - y_1 \\ x_2 - y_2 \end{bmatrix}$$

$$a \mathbf{x} = a \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} ax_1 \\ ax_2 \end{bmatrix}$$

$$\begin{aligned} \mathbf{x} \cdot \mathbf{y} &= \mathbf{x}^T \mathbf{y} = \|\mathbf{x}\| \|\mathbf{y}\| \cos \theta \\ &= [x_1 \quad x_2] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = x_1 y_1 + x_2 y_2 \end{aligned}$$



$$\begin{aligned} \mathbf{W}\mathbf{x} &= \begin{bmatrix} w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \begin{bmatrix} w_{1,1}x_1 + w_{1,2}x_2 + w_{1,3}x_3 \\ w_{2,1}x_1 + w_{2,2}x_2 + w_{2,3}x_3 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{x}^T \mathbf{W} &= [x_1 \quad x_2] \begin{bmatrix} w_{1,1} & w_{1,2} & w_{1,3} \\ w_{2,1} & w_{2,2} & w_{2,3} \end{bmatrix} \\ &= [(x_1 w_{1,1} + x_2 w_{2,1}) \quad (x_1 w_{1,2} + x_2 w_{2,2}) \quad (x_1 w_{1,3} + x_2 w_{2,3})] \end{aligned}$$

$$\mathbf{A}_{3 \times 3} \mathbf{B}_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \times \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} = \begin{bmatrix} a_{11} \times b_{11} + a_{12} \times b_{21} + a_{13} \times b_{31} & a_{11} \times b_{12} + a_{12} \times b_{22} + a_{13} \times b_{32} & a_{11} \times b_{13} + a_{12} \times b_{23} + a_{13} \times b_{33} \\ a_{21} \times b_{11} + a_{22} \times b_{21} + a_{23} \times b_{31} & a_{21} \times b_{12} + a_{22} \times b_{22} + a_{23} \times b_{32} & a_{21} \times b_{13} + a_{22} \times b_{23} + a_{23} \times b_{33} \\ a_{31} \times b_{11} + a_{32} \times b_{21} + a_{33} \times b_{31} & a_{31} \times b_{12} + a_{32} \times b_{22} + a_{33} \times b_{32} & a_{31} \times b_{13} + a_{32} \times b_{23} + a_{33} \times b_{33} \end{bmatrix}$$

Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse:** $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$

```
import numpy as np
```

```
# Calculate inverse
```

```
A_inv = np.linalg.inv(A)
```

- $\det(\mathbf{A})$: determinant of $\mathbf{A}$

$$\det(\mathbf{A}) = \sum_{j=1}^k a_{ij} \mathbf{C}_{ij}, \text{ where cofactor } \mathbf{C}_{ij} = (-1)^{i+j} \mathbf{M}_{ij}$$

$\mathbf{M}_{ij}$: minor of a_{ij} is the determinant obtained by deleting the row and column in which a_{ij} lies.

e.g.,

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$\mathbf{M}_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22} \times a_{33} - a_{23} \times a_{32}$

$\mathbf{C}_{11} = (-1)^{1+1} \mathbf{M}_{11}$

$$\det(\mathbf{A}) = \sum_{j=1}^3 a_{ij} \mathbf{C}_{ij} = a_{i1} \mathbf{C}_{i1} + a_{i2} \mathbf{C}_{i2} + a_{i3} \mathbf{C}_{i3}$$

-Matrix inverse: $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$

- $$\text{adj}(\mathbf{A}) = \mathbf{C}^T, \text{ where cofactor } \mathbf{C}_{ij} = (-1)^{i+j} \mathbf{M}_{ij}$$

$$\text{e.g., } \mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \mathbf{M}_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = a_{22} \times a_{33} - a_{23} \times a_{32}$$

$$\mathbf{C}_{11} = (-1)^{1+1} \mathbf{M}_{11}$$

$$\text{adj}(\mathbf{A})=\mathbf{C}^T = \begin{bmatrix} \mathbf{M}_{11} & -\mathbf{M}_{21} & \mathbf{M}_{31} \\ -\mathbf{M}_{12} & \mathbf{M}_{22} & \mathbf{M}_{32} \\ \mathbf{M}_{13} & -\mathbf{M}_{23} & \mathbf{M}_{33} \end{bmatrix}$$

Review of Fundamentals

❖ Operations on Vectors and Matrices

-Matrix inverse: $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$

- $\det(\mathbf{A}) = \sum_{j=1}^k a_{ij} \mathbf{C}_{ij}$, where cofactor $\mathbf{C}_{ij} = (-1)^{i+j} \mathbf{M}_{ij}$
- $\text{adj}(\mathbf{A}) = \mathbf{C}^T$

$\mathbf{M}_{ij}$: minor of a_{ij} is the determinant obtained by deleting the row and column in which a_{ij} lies.

e.g., What is the cofactor matrix of $\mathbf{A} = \begin{bmatrix} 0 & 1 & 2 \\ 3 & 4 & 5 \\ 6 & 7 & 8 \end{bmatrix}$ $\mathbf{C} = \begin{bmatrix} -3 & 6 & -3 \\ 6 & -12 & 6 \\ -3 & 6 & -3 \end{bmatrix}$

$$\mathbf{C}_{11} = (-1)^{1+1} \begin{vmatrix} 4 & 5 \\ 7 & 8 \end{vmatrix} = -3, \quad \mathbf{C}_{12} = (-1)^{1+2} \begin{vmatrix} 3 & 5 \\ 6 & 8 \end{vmatrix} = 6, \quad \mathbf{C}_{13} = (-1)^{1+3} \begin{vmatrix} 3 & 4 \\ 6 & 7 \end{vmatrix} = -3$$

$$\mathbf{C}_{21} = (-1)^{2+1} \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix} = 6, \quad \mathbf{C}_{22} = (-1)^{2+2} \begin{vmatrix} 0 & 2 \\ 6 & 8 \end{vmatrix} = -12, \quad \mathbf{C}_{23} = (-1)^{2+3} \begin{vmatrix} 0 & 1 \\ 6 & 7 \end{vmatrix} = 6,$$

$$\mathbf{C}_{31} = (-1)^{3+1} \begin{vmatrix} 1 & 2 \\ 4 & 5 \end{vmatrix} = -3, \quad \mathbf{C}_{32} = (-1)^{3+2} \begin{vmatrix} 0 & 2 \\ 3 & 5 \end{vmatrix} = 6, \quad \mathbf{C}_{33} = (-1)^{3+3} \begin{vmatrix} 0 & 1 \\ 3 & 4 \end{vmatrix} = -3,$$

Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse:** $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$

Matrix inverse $\mathbf{A}^{-1}$ exists if and only if the matrix $\mathbf{A}$ is **square and nonsingular** (i.e., **invertible**)

In other words, there exists a square matrix $\mathbf{B}$, such that **$\mathbf{AB}=\mathbf{BA}=\mathbf{I}$** (Identity matrix)

$$\mathbf{I}_{d \times d} = \begin{bmatrix} 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ \vdots & 0 & \ddots & 0 & \vdots \\ 0 & \vdots & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{bmatrix}$$

Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse:** $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$

Matrix inverse $\mathbf{A}^{-1}$ exists if and only if the matrix $\mathbf{A}$ is **square and nonsingular** (i.e., **invertible**)

2 methods to check invertibility

$\det(\mathbf{A})=0?$

$\text{rank}(\mathbf{A}_{d \times d})=d?$

$\text{rank}(\mathbf{A})$: maximal number of **linearly independent** columns/rows of $\mathbf{A}$

$$\beta_1 \mathbf{a}_1 + \beta_2 \mathbf{a}_2 + \cdots + \beta_m \mathbf{a}_m = 0 \text{ only holds for } \beta_1 = \cdots = \beta_m = 0$$

Compute $\text{rank}(\mathbf{A})$: Step 1. Transform into row echelon form using elementary row operations

Step 2. The number of non-zero rows in the row echelon form.

<https://stattrek.com/matrix-algebra/echelon-transform#MatrixA>

Review of Fundamentals

❖ Operations on Vectors and Matrices

-**Matrix inverse:** $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \text{adj}(\mathbf{A})$

Matrix inverse $\mathbf{A}^{-1}$ exists if and only if the matrix $\mathbf{A}$ is **square and nonsingular** (i.e., **invertible**)

e.g., $\begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$

$$\begin{bmatrix} 1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 9 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 0 & 9 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 1 \\ 3 & 4 & 5 \end{bmatrix}$$

$$2 \times R_2 + R_1 = R_3$$

Review of Fundamentals

❖ Some properties on Matrix Operations

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

$$(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$$

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$$

$$(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$$

$$\mathbf{I}_m \mathbf{A}_{m \times n} = \mathbf{A} = \mathbf{A}_{m \times n} \mathbf{I}_n$$

$$(\mathbf{A}^T)^T = \mathbf{A}$$

$$(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$$

$$(r\mathbf{A})^T = r\mathbf{A}^T$$

$$(\mathbf{AB})^T = \mathbf{B}^T \mathbf{A}^T$$

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$$

$$(r\mathbf{A})^{-1} = r^{-1}\mathbf{A}^{-1}$$

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

$$\mathbf{I}^{-1} = \mathbf{I}$$

$$(\mathbf{A}^T)^{-1} = (\mathbf{A}^{-1})^T$$

$$(\mathbf{A}^{-1})^{-1} = \mathbf{A}$$

Review of Fundamentals

❖ Sets, Functions

-A **Set** is an unordered collection of unique elements

Denoted by a calligraphic capital letter (e.g., $\mathcal{X}$, $\mathcal{Y}$)

$x \in \mathcal{X}$: the element x belongs to the set $\mathcal{X}$

✓ Finite set: a fixed number of elements

Denoted by accolades (e.g., $\{1, 3.5, 7, -1\}$)

✓ Infinite set: include all values in some interval

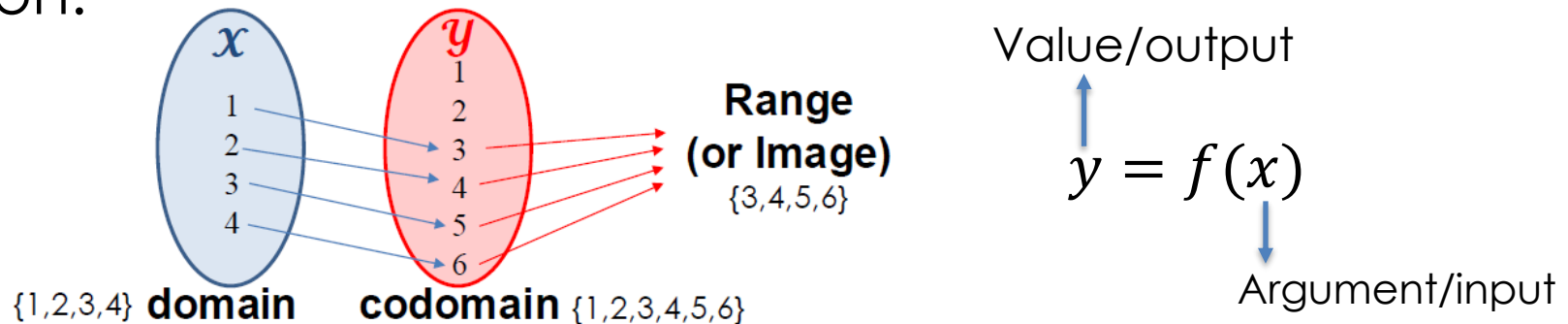
Denoted by square brackets as $[a, b]$ if including a and b
(e.g., $[1, 2]$)

Denoted by parentheses as (a, b) if does not include a and b
(e.g., $(1, 2)$)

Review of Fundamentals

❖ Sets, Functions

-A **Function** is a relation that associates each element x of a set $\mathcal{X}$, the domain of the function, to a single element y of another set $\mathcal{Y}$, the codomain of the function.



- ✓ Scaler function: function returns a scalar (e.g., $f(x)$ or $f(\mathbf{x})$)
- ✓ Vector function: function returns a vector (e.g., $\mathbf{f}(x)$ or $\mathbf{f}(\mathbf{x})$)

Review of Fundamentals

❖ Operations on Functions

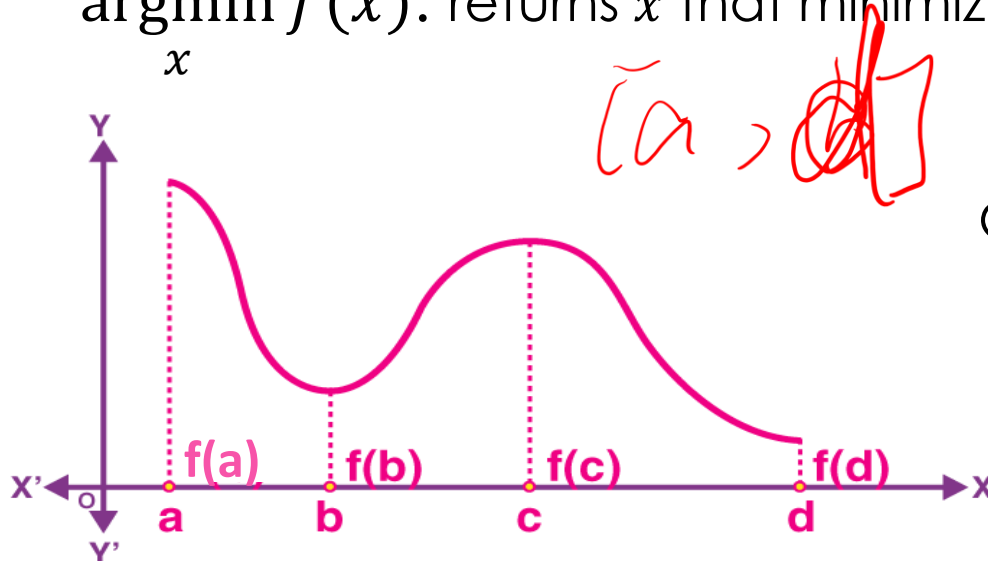
-Maximum and Minimum

$\max_x f(x)$: returns the highest value of $f(x)$ across its domain

$\operatorname{argmax}_x f(x)$: returns x that maximizes $f(x)$

$\min_x f(x)$: returns the lowest value of $f(x)$ across its domain

$\operatorname{argmin}_x f(x)$: returns x that minimizes $f(x)$



Global maximum = $\operatorname{argmax}_x f(x)$

Global minimum = $\operatorname{argmin}_x f(x)$

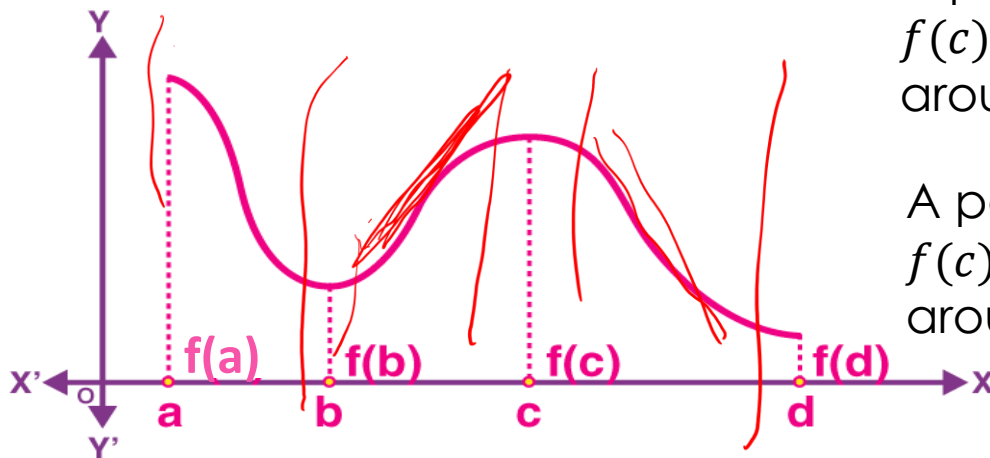
Review of Fundamentals

❖ Operations on Functions

-Derivative (single-input)

The derivative f' of a function f is a function that describes how fast f grows (or decreases)

- If f' is positive at some x , f grows at this x
- If f' is negative at some x , f decreases at this x
- If f' is zero at some x , f 's slope at x is horizontal



A point $x=c$ is a **local maximum** of $f(x)$ if $f(c) \geq f(x)$ for all x in some **open** interval around c

A point $x=c$ is a **local minimum** of $f(x)$ if $f(c) \leq f(x)$ for all x in some **open** interval around c

Review of Fundamentals

❖ Operations on Functions

-Gradient (multiple-input scalar function)

Generalization of derivative for scalar functions with multiple inputs.

Denoted by $\nabla_{\mathbf{x}}f$

A vector of **partial derivatives**

$$\nabla_{\mathbf{x}}f = \frac{df(\mathbf{x})}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_d} \end{bmatrix}$$

Differentiation of $f(\mathbf{x})$ w.r.t. $\mathbf{x}$

e.g., $f(\mathbf{x}) = 2x_1 + 3x_2$, what is $\nabla_{\mathbf{x}}f$?

A matrix of **partial derivatives**

$$\nabla_{\mathbf{X}}f = \frac{df(\mathbf{X})}{d\mathbf{X}} = \begin{bmatrix} \frac{\partial f}{\partial x_{1,1}} & \cdots & \frac{\partial f}{\partial x_{1,K}} \\ \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial x_{d,1}} & \cdots & \frac{\partial f}{\partial x_{d,K}} \end{bmatrix}$$

same size as $\mathbf{X}$

$$\begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

Review of Fundamentals

❖ Operations on Functions

-Jacobian (multiple-input vector function)

Generalization of derivative for vector functions with multiple inputs.

A matrix of **partial derivatives**

$$\begin{bmatrix} 2 & 3 \\ 5x_2 & x_1 \end{bmatrix}$$

$$\frac{d\mathbf{f}(\mathbf{x})}{d\mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \dots & \frac{\partial f_1}{\partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_h}{\partial x_1} & \dots & \frac{\partial f_h}{\partial x_d} \end{bmatrix}$$

$$\frac{\partial (2x_1 + 3x_2)}{\partial x_1}$$
$$\frac{\partial (5x_1x_2)}{\partial x_1}$$

Differentiation of $\mathbf{f}(\mathbf{x})$ of size $h \times 1$ w.r.t. $\mathbf{x}$ of size $d \times 1$.

e.g., $\mathbf{f}(\mathbf{x}) = \begin{bmatrix} 2x_1 + 3x_2 \\ 5x_1x_2 \end{bmatrix}$, what is the Jacobian matrix $\frac{d\mathbf{f}(\mathbf{x})}{d\mathbf{x}}$?

Review of Fundamentals

❖ Common Differentiation Rules

- Constant Rule: the derivative of a constant is 0
- Sum & Difference Rule: $(f(x) + g(x))' = f'(x) + g'(x)$
 $(f(x) - g(x))' = f'(x) - g'(x)$
- Product Rule: $(f(x)g(x))' = f(x)g'(x) + f'(x)g(x)$
- Quotient Rule: $\left(\frac{f(x)}{g(x)}\right)' = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$
- Chain Rule: $(f(g(x)))' = f'(g(x))g'(x)$ or $\frac{df}{dx} = \frac{df}{dg} \frac{dg}{dx}$

THE END